Department of Computer Science
Machine Learning Project Proposal
Project Topic: Human Activity Detection

Section: D
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Explanation: This project is a **supervised learning image classification system**. The model is trained on labeled images of different human activities such as **calling, clapping, cycling, and sleeping etc**. Each image belongs to a specific class, which helps the model learn patterns associated with each activity.

First, the dataset is preprocessed by resizing images, normalizing pixel values, and applying data augmentation techniques to improve model generalization. Then, a pre-trained **MobileNetV2** model is used through transfer learning, where existing learned features are reused and fine-tuned for this specific task.

During training, the model learns important visual features such as body posture, movement patterns, and object interaction. After training, when a new unseen image is provided, the model analyzes it and predicts the most likely human activity along with a confidence score.